

Particle Swarm Optimization and the Fully Informed Particle Swarm

A Comparative Analysis

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- 1 Introduction
- 2 Optimization and Swarm Intelligence
- 3 Canonical PSO
- 4 PSO vs. FIPS
- 5 Numerical Simulations
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“That which is not good for the swarm, neither is it good for the bee.”
— Marcus Aurelius, *Meditations*

- Identifying the optimal configuration of a system is fundamentally a **search problem**.
- Classical derivative-based methods fail when the objective function is non-differentiable, discontinuous, or analytically unavailable.
- **PSO** (Kennedy & Eberhart, 1995) is a derivative-free metaheuristic in which a population of interacting particles collectively explores the search space.

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5 principles of swarm systems (*Millonas*)

- Proximity, Quality, Diverse response
- **Stability vs. Adaptability**

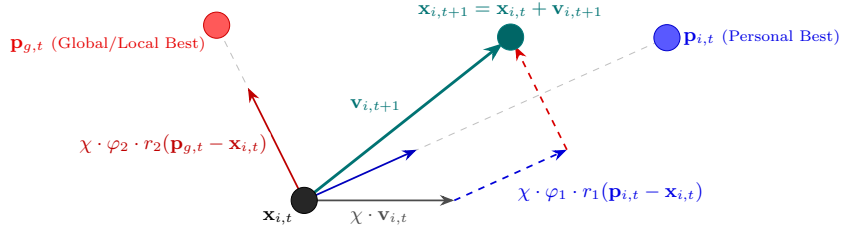
This stability/adaptability trade-off anticipates the comparison between PSO and FIPS developed in Chapter 4.

Two direct sources of PSO

- **Boid model** (Reynolds, 1987)
- **Cornfield model** (Heppner)
- Kennedy & Eberhart (1995) formalized these ideas into the original PSO update rule.

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Velocity and Position Update



Vector decomposition of the update (adapted from the thesis, Fig. 3.1); here χ replaces ω and φ_1, φ_2 replace c_1, c_2 in the constriction-factor formulation used throughout this thesis:

$$v_i^d(t+1) = \chi \left[v_i^d(t) + \varphi_1 r_1 (p_i^d - x_i^d(t)) + \varphi_2 r_2 (p_g^d - x_i^d(t)) \right]$$

- **Cognitive term** (φ_1): attracts the particle toward its own personal best p_i , representing its individual experience.
- **Social term** (φ_2): attracts the particle toward the swarm's best p_g , representing social cooperation.
- Higher $\varphi_1/\varphi_2 \Rightarrow$ conservative behavior, with greater emphasis on the particle's own experience.
- Higher $\varphi_2/\varphi_1 \Rightarrow$ faster but riskier convergence toward the swarm best.

Historical evolution: fixed coefficients (Kennedy & Eberhart, 1995) $\rightarrow$ inertia weight ω (Shi & Eberhart, 1998) $\rightarrow$ constriction factor χ (Clerc & Kennedy, 2002).

The Constriction Factor

- Controls the dynamics of particle trajectories and provides stability without the need for explicit velocity clamping (*Clerc & Kennedy, 2002*).
- It is derived from a stability analysis of the swarm as a discrete-time linear dynamical system, with the eigenvalues constrained within the unit circle.

$$\chi = \begin{cases} \frac{2\kappa}{\left| 2 - \varphi - \sqrt{\varphi^2 - 4\varphi} \right|} & \varphi = \varphi_1 + \varphi_2 > 4 \\ \kappa & \text{otherwise} \end{cases}$$

- **Thesis setting:** $\kappa = 1$, $\varphi_1 = \varphi_2 = 2.05 \Rightarrow \varphi = 4.1$, $\chi \approx 0.7298$.

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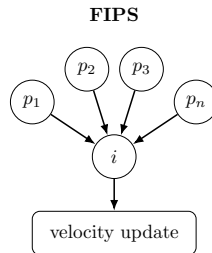
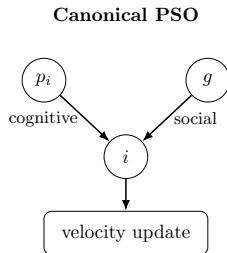
From PSO to FIPS: Motivation

Limitation of canonical PSO

The social component is based on a *single* social reference (the best-performing neighbor) discarding the experience of every other particle in the neighborhood.

Fully Informed Particle Swarm (*Mendes, Kennedy & Neves, 2004*)

Every neighbor contributes to the velocity update, not just the best-performing one.



FIPS: Formulation (Unweighted, Self-Inclusive)

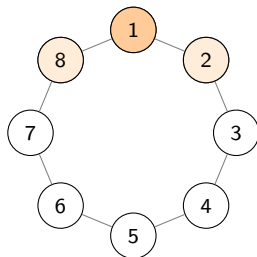
$$v_{i,t+1}^d = \chi \left(v_{i,t}^d + \varphi \left(p_{m,i,t}^d - x_{i,t}^d \right) \right), \quad p_{m,i,t}^d = \frac{\sum_{k \in N_i} W(k) \varphi_k^d p_{k,t}^d}{\sum_{k \in N_i} W(k) \varphi_k^d}$$

$$\varphi_k^d = U \left[0, \frac{\varphi}{|N_i|} \right] \quad \forall k \in N_i$$

- N_i : neighborhood of particle i (here **including** i itself)
- p_k : personal best of neighbor k ; unweighted $\Rightarrow W(k) = 1$ for all k
- **Thesis setting:** $\varphi = 4.1$ (same total coefficient as canonical PSO), $\chi \approx 0.7298$
- Canonical PSO = special case with $|N_i| = 2$, weighted by fixed φ_1, φ_2

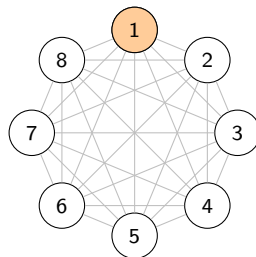
Neighborhood Topologies

Ring ($K = 3$)



Each particle is connected only to its two immediate neighbors.

All ($K = N$)



Every particle is connected to every other particle.

PSO vs. FIPS: Analogies and Differences

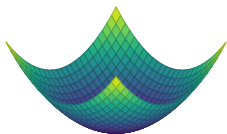
	Canonical PSO	FIPS
Social information source	Single best neighbor	Weighted average of all neighbors
Effect of larger neighborhood	More candidates from which to pick the best	More (potentially conflicting) contributions
Weighting function $W(k)$	Not applicable	Determines the relative influence of each neighbor
Convergence mechanism	Constriction factor χ	Same χ , generalized to K sources

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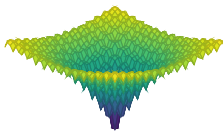
Benchmark Functions

Sphere



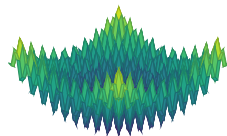
Unimodal, convex

Ackley



Nearly flat, deep central basin,
many small local optima

Rastrigin



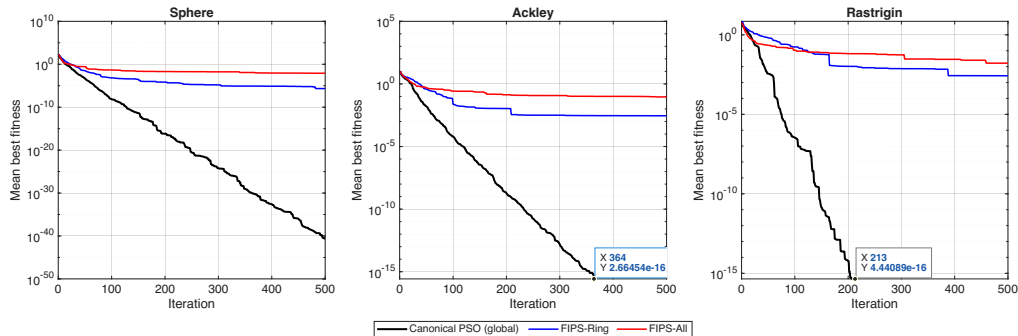
Highly multimodal, regularly spaced
local optima

All three functions have global minimum $f(0,0) = 0$. Actual search domains: Sphere $[-100, 100]^2$, Ackley $[-32, 32]^2$, Rastrigin $[-5.12, 5.12]^2$ (plots above use a common scale for visual comparison).

Parameter	Value
Swarm size N	30
Dimensionality D	2
Independent runs	20 per configuration
Max. iterations	500
Boundary handling	Reflecting wall
Success tolerance δ	$0.0015 \times (ub - lb)$
Tight tolerance δ_{tight}	$10^{-4} \times (ub - lb)$

Same random initial swarm (X_0, V_0) shared across all three configurations, per run. Tolerances follow Lange, Schmitt & Wanka (2014), calibrated for low-dimensional search spaces ($D = 2$) and scaled to the domain width.

Results: Convergence Behavior



- Canonical PSO decreases sharply and almost monotonically on all three functions, reaching machine precision well before iteration 500.
- Both FIPS configurations decrease for ~ 30 – 50 iterations, then **flatten out** — an early qualitative indication of stagnation.

Results: Final Fitness, Success Rate & Stagnation

Function	Config.	Mean best	SR (δ)	SR (δ_{tight})	Med. iters (δ_{tight})
Sphere	Can. PSO	2.63×10^{-41}	100%	100%	44.5
	FIPS-Ring	2.18×10^{-6}	100%	100%	80.0
	FIPS-All	8.16×10^{-3}	100%	20%	363.5
Ackley	Can. PSO	≈ 0	100%	100%	47.0
	FIPS-Ring	2.86×10^{-3}	100%	95%	70.0
	FIPS-All	9.12×10^{-2}	100%	20%	82.0
Rastrigin	Can. PSO	0	100%	100%	55.5
	FIPS-Ring	2.67×10^{-3}	100%	85%	185.0
	FIPS-All	1.66×10^{-2}	95%	35%	122.0

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The Core Argument, in Three Sentences

- ① More information sources do not necessarily lead to a better-performing swarm.
- ② Neighborhood topology is not merely a technical detail—it affects the resulting search dynamics.
- ③ Similar fitness values can correspond to substantially different positions: therefore, an appropriate convergence criterion is essential.

Limitations

- Restricted to $D = 2$
- Only unweighted FIPS, Ring & All topologies
- Symmetric initialization (vs. Mendes' asymmetric design)

Future work

- Extend the analysis to higher dimensionality
- Investigate weighted FIPS variants & more topologies
- Introduce formal statistical significance tests

Thank you for your attention

Backup Slides

for additional questions

Constriction factor: full derivation

Full results table

5 topologies

FIPS variants

All limitations

Backup: Constriction Factor — Derivation (Clerc & Kennedy, 2002, §IV-C)

Model (Eq. 4.8–4.9): $v(t+1) = \chi(v(t) + \varphi(p - x(t)))$, $x(t+1) = v(t+1) + x(t)$.

Step 1) Matrix of the System.

$$M = \begin{pmatrix} \chi & \chi\varphi \\ -\chi & 1 - \chi\varphi \end{pmatrix}$$

Step 2) Eigenvalues. Roots of $Z^2 - \text{tr}(M)Z + \det(M) = 0$:

$$Z^2 - (\chi + 1 - \chi\varphi)Z + \chi = 0, \quad e_{1,2} = \frac{(\chi + 1 - \chi\varphi) \pm \sqrt{\Delta}}{2}$$

Step 3) Complex and Real Areas on φ . $\Delta < 0$ on an interval of φ (depending on χ) $\Rightarrow e_{1,2}$ are true complex numbers, with common modulus $|e| = \sqrt{\chi}$.

Step 4) Extension of the Complex Region & Constriction Coefficient. In the complex region, the convergence criterion requires $\chi < 1$. Trying $\chi = \kappa/|e_2|$, $\kappa \in]0, 1[$, gives

$$\chi = \begin{cases} \frac{2\kappa}{\varphi - 2 + \sqrt{\varphi^2 - 4\varphi}} & \varphi > 4 \\ \kappa & \text{otherwise} \end{cases}$$

Thesis setting: $\kappa = 1$, $\varphi = \varphi_1 + \varphi_2 = 4.1 > 4 \Rightarrow \chi \approx 0.7298$.

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Backup: Full Numerical Results (Table 5.1)

Func.	Config.	Mean best	Med. best	Std	SR	t_{hit}	SR^T	t_{hit}^T
Sphere	Can. PSO	2.63e-41	2.88e-43	9.00e-41	100%	23.5	100%	44.5
	FIPS-Ring	2.18e-6	5.59e-8	5.52e-6	100%	36.0	100%	80.0
	FIPS-All	8.16e-3	2.07e-3	1.27e-2	100%	38.0	20%	363.5
Ackley	Can. PSO	≈ 0	≈ 0	0	100%	23.0	100%	47.0
	FIPS-Ring	2.86e-3	5.41e-5	9.85e-3	100%	39.0	95%	70.0
	FIPS-All	9.12e-2	7.46e-2	7.40e-2	100%	31.5	20%	82.0
Rastrigin	Can. PSO	0	0	0	100%	32.0	100%	55.5
	FIPS-Ring	2.67e-3	3.06e-8	1.03e-2	100%	79.0	85%	185.0
	FIPS-All	1.66e-2	1.52e-3	6.03e-2	95%	64.0	35%	122.0

$N_{runs} = 20$. SR/SR^T : success rate under δ/δ_{tight} . t_{hit} : median iterations to first success (successful runs only).
Ackley Can. PSO reported as -4.441×10^{-16} : machine precision, numerically $\equiv 0$.

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Backup: The Five Neighborhood Topologies (Mendes et al.)

Topology	Avg. Distance	Diameter	Distribution Sequence
All	1.00	1	$\langle 19 \rangle$
Ring	5.26	10	$\langle 2, 2, 2, 2, 2, 2, 2, 2, 2, 1 \rangle$
Four Clusters	2.26	3	$\langle 4.6, 4.8, 9.6 \rangle$
Pyramid	2.04	4	$\langle 5.4, 7.8, 5.4, 0.4 \rangle$
Square	2.32	4	$\langle 4, 7, 6, 2 \rangle$

Statistics measured on $N = 20$ particles (Mendes et al., 2004). Ring has the largest diameter (slowest information spread); All has diameter 1 (instant, full spread) but performed poorly in their experiments — consistent with the FIPS-All stagnation observed in this thesis.

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Backup: FIPS Variants and Self-Inclusion

- **FIPS** (used in this thesis): unweighted, $W(k) = 1$ for all neighbors.
- **wFIPS**: $W(k) \propto$ fitness of neighbor k 's personal best.
- **wdFIPS**: $W(k) \propto$ distance to the particle being updated.
- **Self** / **wSelf**: the particle's own best is explicitly included in N_i , receiving half the total weight by construction.

Mendes et al. found several *self-excluding* (“U-”) topologies performed as well as or better than their self-including counterparts — suggesting a particle's own history may already be reflected in its current position and velocity. This thesis uses only self-inclusive, unweighted FIPS (Ring, All).

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Backup: Complete List of Limitations (§6.2)

- Restricted to a two-dimensional search space
- Only unweighted FIPS; only Ring and All topologies tested
- Symmetric initialization (vs. Mendes' asymmetric design)
- Descriptive statistics only — no formal hypothesis testing
- Extra benchmark functions (Griewank, Rosenbrock, Schaffer f6) shown but not used in the comparison
- No velocity clamping (constriction factor only; not compared against inertia weight + clamping)
- Only self-inclusive FIPS configurations (no self-excluding “U-” variants)
- Tolerance thresholds $\delta, \delta_{\text{tight}}$ not systematically sensitivity-tested

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- D. Wang, D. Tan, L. Liu, "Particle Swarm Optimization Algorithm: An Overview," *Soft Computing*, vol. 22, 2017.
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- V. Lange, M. Schmitt, R. Wanka, "Towards a Better Understanding of the Local Attractor in Particle Swarm Optimization: Speed and Solution Quality," Springer, 2014.